

Supply and Demand

Chapter 3

ECON 1450: Principles of Microeconomics

University of Vermont

2026-01-01

Supply and Demand

Chapter 3

ECON 1450: Principles of Microeconomics

University of Vermont ■ Fall 2026

How much for a hotel room in Burlington on April 7, 2024?

Typical spring weekend: - Comfort Inn: \$120 - Hilton Garden Inn: \$150

Eclipse weekend (April 7-8, 2024): - Comfort Inn: \$475 - Hilton Garden Inn: \$860 - Many hotels: fully booked

What changed?

Same hotels, same rooms, same city... dramatically different prices. **Supply and demand in action.**



Image: Burlington during 2024 total solar eclipse

By October 2023, the picture was clear

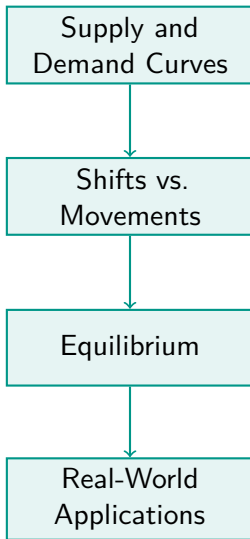
Accommodation listings across Vermont were being booked months in advance:

- La Quinta Inn & Suites: \$624/night
- Best Western Plus: \$475/night
- Hilton Garden Inn: \$860/night
- VRBO and Airbnb: elevated pricing throughout the region

Market Response

Hotels didn't build new rooms. They didn't change amenities. They simply adjusted the price to match the surge in demand from eclipse viewers.

Today's Journey



Learning Objectives

By the end of today's class, you will be able to:

- ➊ **Define** competitive markets and explain how supply and demand describe them
- ➋ **Distinguish** between movements along curves versus shifts of curves
- ➌ **Identify** the five factors that shift demand and five that shift supply
- ➍ **Find** market equilibrium and explain how prices eliminate shortages and surpluses
- ➎ **Predict** how price and quantity change when markets experience shocks

Core Skill

The supply-demand framework is economics' most powerful analytical tool. Master it, and you can explain everything from hotel prices to labor markets to cryptocurrency.

Competitive Markets: Our Starting Point

For now, we analyze **competitive markets**:

Characteristics:

- Many buyers and sellers
- No single participant can influence market price
- Standardized products
- Free entry and exit

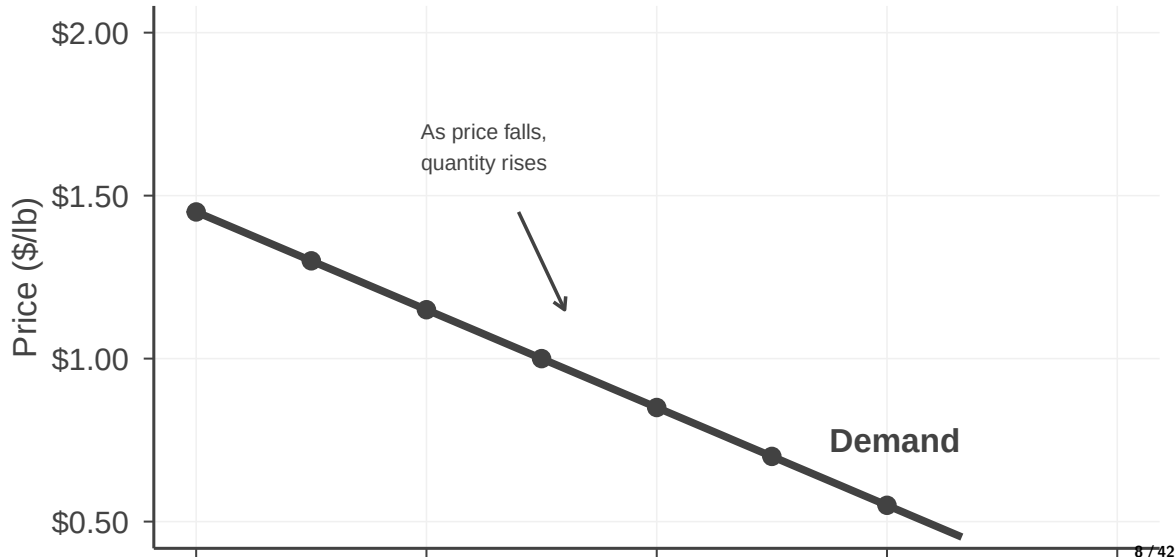
Examples:

- Agricultural commodities (wheat, corn)
- Stock markets
- Online marketplaces
- Gasoline stations in a city

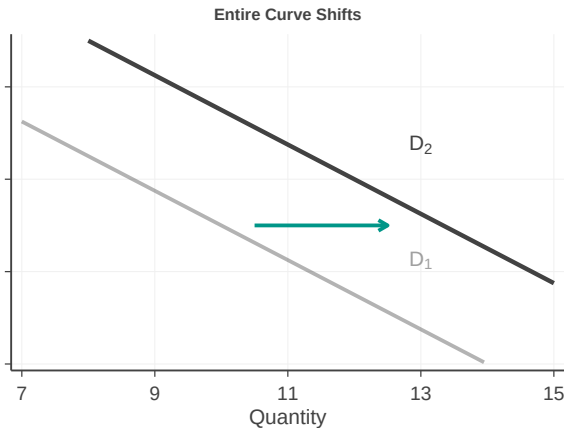
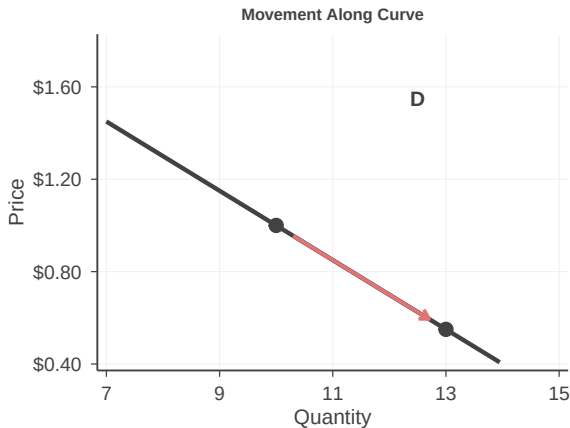
Why start here?

Not all markets are competitive (we'll study monopolies and oligopolies later), but competitive markets provide the clearest insights into how prices coordinate economic activity.

The Demand Curve



Critical Distinction: Movement vs. Shift



Pro Tip: If the changing variable is on the Y-axis (price), it's a **movement**. Otherwise, it's a **shift**!

Common Mistakes: Movement vs. Shift

Students often confuse these concepts. Here's how to avoid errors:

WRONG: “Price increased, so demand shifted” - **CORRECT:** “Price increased, so quantity demanded decreased” (movement along curve)

WRONG: “Income rose, so quantity demanded increased” - **CORRECT:** “Income rose, so demand increased” (entire curve shifts)

Quick Test

Ask yourself: “Did **price** change, or did **something else** change?” - Price changed → **movement** along the curve - Something else changed → **shift** of the entire curve

What Shifts Demand? The TRIBE Framework

Five Demand Shifters

- T** - Tastes and preferences
- R** - Related goods prices
- I** - Income
- B** - Buyers (number of consumers)
- E** - Expectations

Remember:

These factors shift the **entire curve**

Price changes cause **movements along** the curve

Direction matters: increase vs. decrease in demand

Demand Shifter 1: Related Goods

Substitutes - Goods that can replace each other

- $\uparrow$ Price of Hawaii trips $\rightarrow \uparrow$ Demand for Puerto Rico trips
- $\uparrow$ Price of beef $\rightarrow \uparrow$ Demand for chicken
- $\uparrow$ Price of iPhone $\rightarrow \uparrow$ Demand for Android phones

Price of substitute $\uparrow \rightarrow$ Demand $\uparrow$

Complements - Goods consumed together

- $\uparrow$ Price of cars $\rightarrow \downarrow$ Demand for gasoline
- $\uparrow$ Price of printers $\rightarrow \downarrow$ Demand for ink cartridges
- $\uparrow$ Price of hot dogs $\rightarrow \downarrow$ Demand for hot dog buns

Price of complement $\uparrow \rightarrow$ Demand $\downarrow$



Substitutes: iPhone vs Android

Knowledge Check: Substitutes and Complements

A new tax raises the price of gasoline. What happens to the demand for electric vehicles?

- a)** Demand for electric vehicles increases
- b)** Demand for electric vehicles decreases
- c)** Quantity demanded of electric vehicles increases
- d)** No change in demand for electric vehicles

Knowledge Check: Substitutes and Complements

A new tax raises the price of gasoline. What happens to the demand for electric vehicles?

- ☐ a) Demand for electric vehicles increases
- ☐ b) Demand for electric vehicles decreases
- ☐ c) Quantity demanded of electric vehicles increases
- ☐ d) No change in demand for electric vehicles

∴ .block ****Answer:**** (a) Gasoline and electric vehicles are substitutes. When the price of gasoline rises, consumers shift toward the substitute → demand for EVs increases (curve shifts right). ∴

Demand Shifter 2: Income

Normal Goods

Income $\uparrow \rightarrow$ Demand $\uparrow$

Examples: - Restaurant meals - New cars -
International travel - Organic produce

Most goods are normal goods

Inferior Goods

Income $\uparrow \rightarrow$ Demand $\downarrow$

Examples: - Instant ramen - Used cars (vs. new)
- Generic brands - Bus rides (vs. owning a car)

Budget substitutes we trade up from

Knowledge Check: Inferior Goods

Which are most likely to be inferior goods?

- ☐ a) Store-brand cereal
- ☐ b) Designer boots
- ☐ c) Premium deodorant
- ☐ d) Public bus rides

Knowledge Check: Inferior Goods

Which are most likely to be inferior goods?

- a) Store-brand cereal
- b) Designer boots
- c) Premium deodorant
- d) Public bus rides

∴ .block **Answer:** (a) and (d)

Store-brand cereal and bus rides are budget options that people typically replace with premium alternatives (name-brand cereal, personal vehicles) as income rises. ∴

Demand Shifter 3: Tastes and Preferences

Consumer preferences change for many reasons:

Seasonal patterns - Winter $\rightarrow$ $\uparrow$ demand for coats, snow tires - Summer $\rightarrow$ $\uparrow$ demand for air conditioners, beach gear

Trends and fads - Stanley cups, fidget spinners, Pokémon cards - These create temporary demand spikes

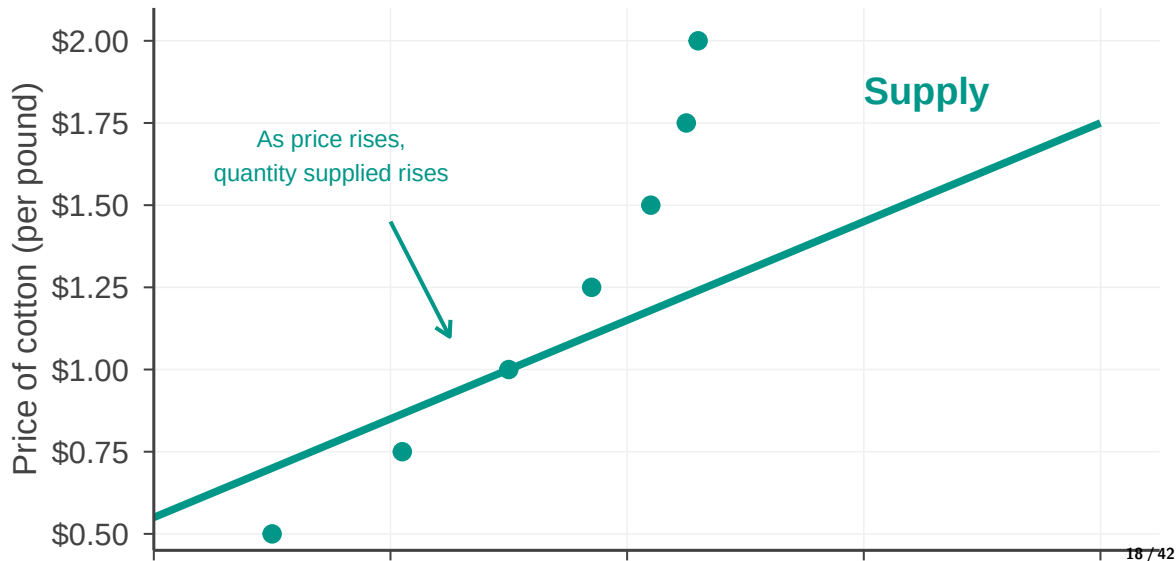
New information - Health studies linking smoking to cancer $\rightarrow$ $\downarrow$ demand for cigarettes - Celebrity endorsements $\rightarrow$ $\uparrow$ demand for products

Demand Shifters 4-5: Expectations and Buyers

4. Expectations - Expect higher future prices $\rightarrow$ $\uparrow$ demand today (buy now!) - Expect lower future prices $\rightarrow$ $\downarrow$ demand today (wait for sale) - Expect higher future income $\rightarrow$ $\uparrow$ demand today

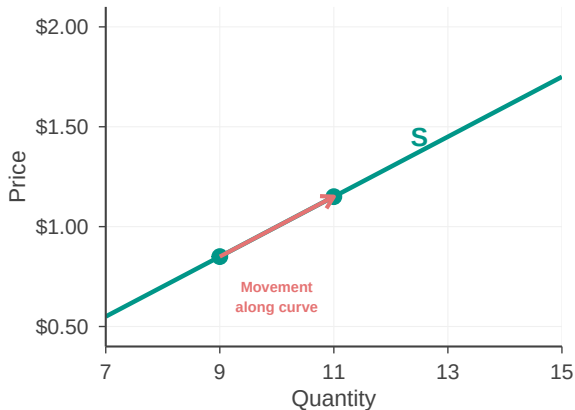
5. Number of Buyers - More consumers $\rightarrow$ $\uparrow$ demand - Population growth, new markets opening - Example: More babies $\rightarrow$ $\uparrow$ demand for diapers

The Supply Curve

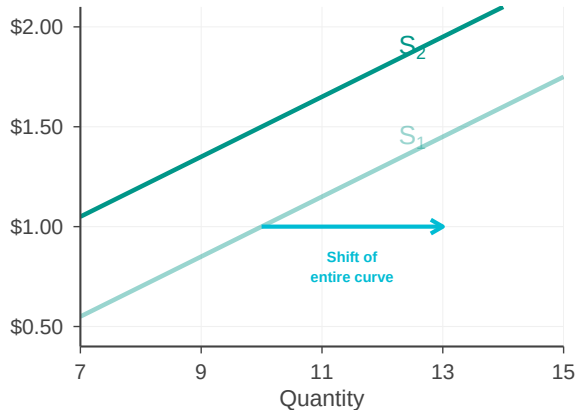


Movement vs. Shift: Supply Edition

Change in Quantity Supplied



Change in Supply



What Shifts Supply? The TONER Framework

Five Supply Shifters

T - Technology

O - Other goods prices

N - Number of producers

E - Expectations

R - Resource costs (inputs)

Key Insight:

Supply shifts when producers' **costs** or **profit opportunities** change

Better technology, more producers, or lower input costs $\rightarrow \uparrow$ supply

Higher input costs or fewer producers $\rightarrow \downarrow$ supply

Supply Shifter 1: Input Prices

↓ Input Price → ↑ Supply

Lower production costs mean producers can profitably supply more at each price

Examples:

- ↓ Fertilizer prices → ↑ Cotton supply
- ↓ Steel prices → ↑ Car supply
- ↓ Microchip prices → ↑ Laptop supply

↑ Input Price → ↓ Supply

Higher production costs reduce profitable supply at each price

Examples:

- ↑ Labor costs → ↓ Restaurant supply
- ↑ Fuel costs → ↓ Trucking supply
- ↑ Coffee bean costs → ↓ Latte supply



Input costs affect production

Supply Shifter 2: Related Goods in Production

Producers often can make multiple products (substitutes) or must make them together (complements)

Substitutes in Production

- Farmer's land: corn vs. soybeans
- $\uparrow$ Price of soybeans $\rightarrow$ farmer plants more soybeans $\rightarrow$ $\downarrow$ corn supply

Complements in Production

- Beef and leather (from same cow)
- Pork and lard (from same pig)
- $\uparrow$ Demand for beef $\rightarrow$ $\uparrow$ beef production $\rightarrow$ $\uparrow$ leather supply

Knowledge Check: Production Relationships

A drought devastates corn crops. Farmers can grow either corn or soybeans on their land. What happens in the soybean market?

- a)** Supply of soybeans increases
- b)** Supply of soybeans decreases
- c)** Demand for soybeans increases
- d)** Quantity supplied of soybeans increases

Knowledge Check: Production Relationships

A drought devastates corn crops. Farmers can grow either corn or soybeans on their land. What happens in the soybean market?

- a)** Supply of soybeans increases
- b)** Supply of soybeans decreases
- c)** Demand for soybeans increases
- d)** Quantity supplied of soybeans increases

∴ .block ****Answer:**** (a) Corn and soybeans are ****substitutes in production****. When corn crops fail, farmers switch their land to soybeans → supply of soybeans increases (shifts right). ∴

Supply Shifter 3: Technology

Better technology lowers production costs:

Examples: - Precision agriculture: GPS-guided tractors → less waste, better yields - Industrial automation: robots → lower labor costs per unit - AI and machine learning: optimized logistics → reduced shipping costs

Key feature: Technology improvements are usually **permanent**

- Once adopted, they become the new standard
- Competitors must adopt or risk being priced out

Result

Better technology → ↓ costs → ↑ supply (shift right)

Supply Shifters 4-5: Expectations and Sellers

4. Expectations - Expect higher future prices $\rightarrow$ $\downarrow$ supply today (save inventory) - Expect lower future prices $\rightarrow$ $\uparrow$ supply today (sell now!) - Example: Oil companies withhold production when expecting price increases

5. Number of Producers - More sellers enter market $\rightarrow$ $\uparrow$ supply - Sellers exit market $\rightarrow$ $\downarrow$ supply - Entry/exit depends on profitability and barriers to entry - Example: Legal marijuana $\rightarrow$ hundreds of new producers entered market $\rightarrow$ $\uparrow$ supply

Knowledge Check: Supply Shifters

Green Mountain Solar knows the government is phasing out solar tax credits in two years. What supply shifter best describes this?

- ☐ a) Input prices
- ☐ b) Prices of related goods
- ☐ c) Technology
- ☐ d) Expectations
- ☐ e) Number of sellers

Knowledge Check: Supply Shifters

Green Mountain Solar knows the government is phasing out solar tax credits in two years. What supply shifter best describes this?

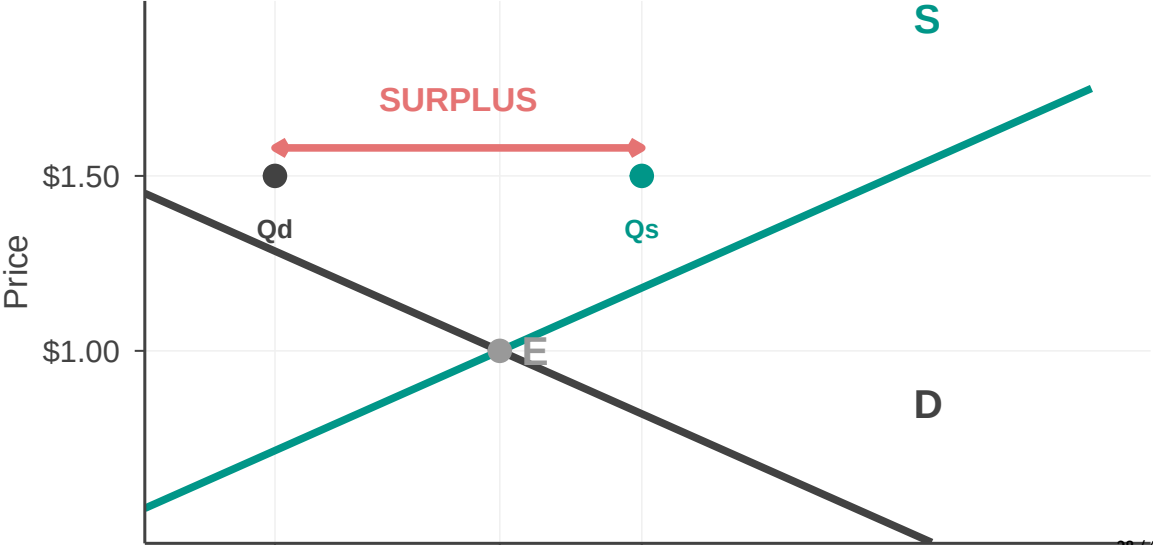
- ☐ a) Input prices
- ☐ b) Prices of related goods
- ☐ c) Technology
- ☐ d) Expectations
- ☐ e) Number of sellers

∴ .block ****Answer:**** (d) Expectations Anticipating less favorable conditions in the future affects production decisions today. ∴

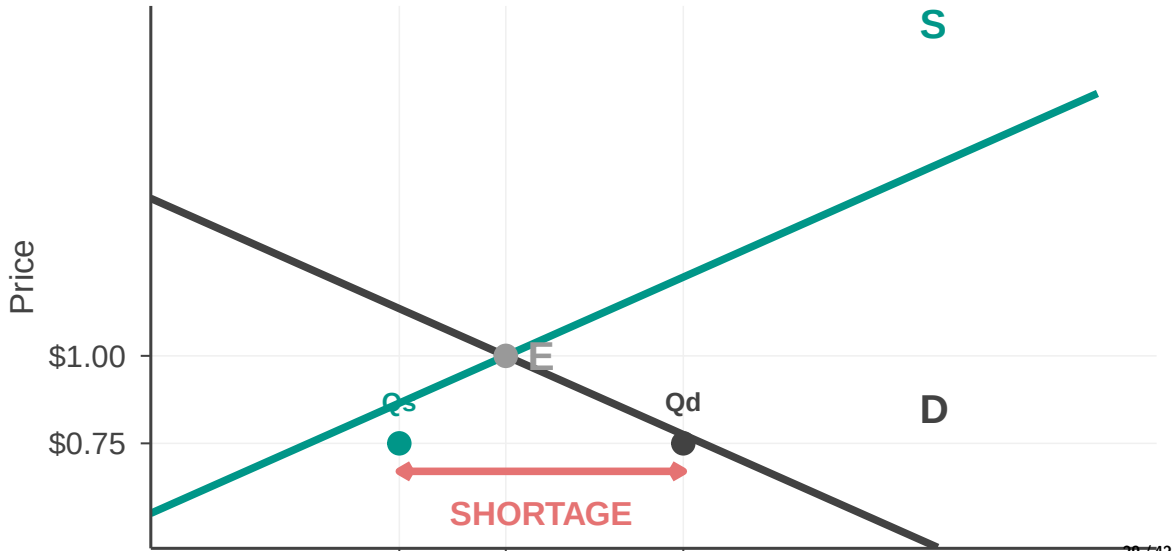
Finding Equilibrium: Where Supply Meets Demand



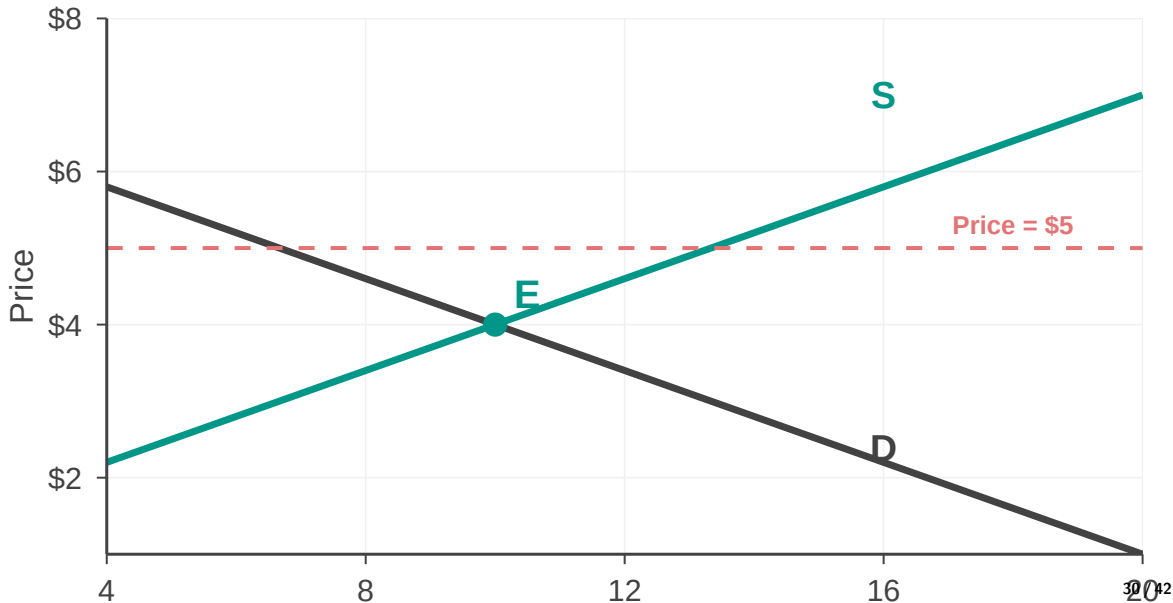
Surplus: When Price is Too High



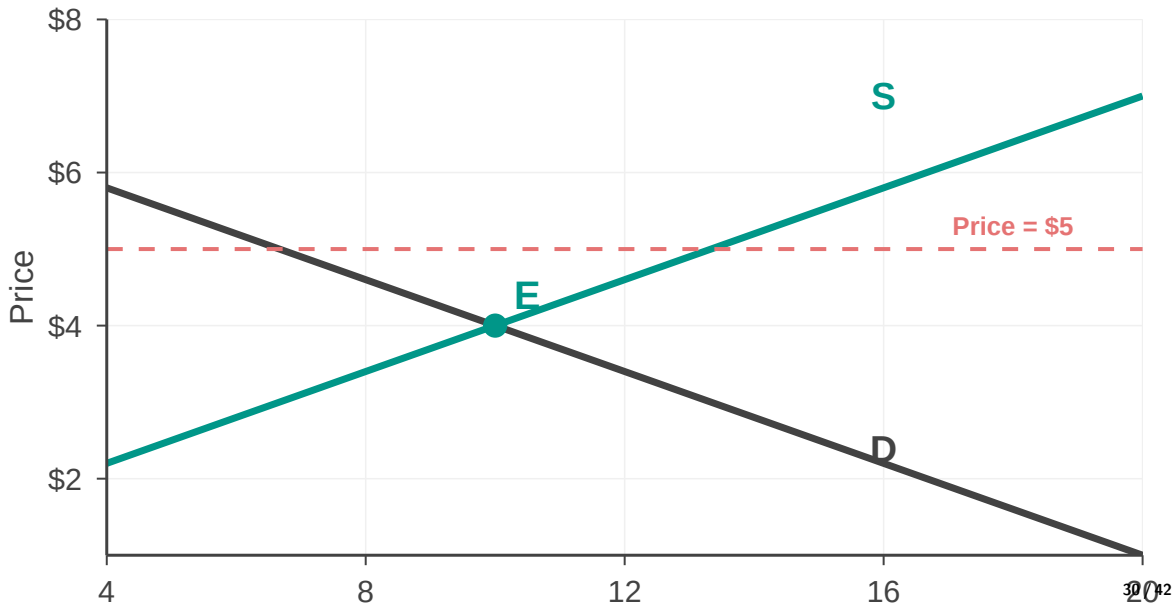
Shortage: When Price is Too Low



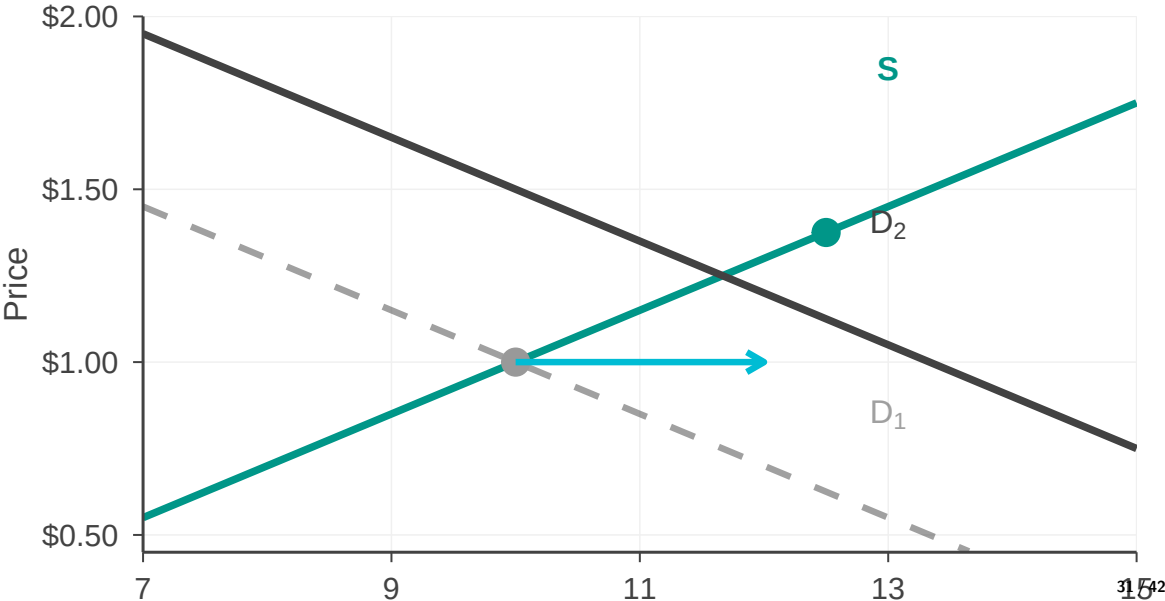
Knowledge Check: Shortage or Surplus?



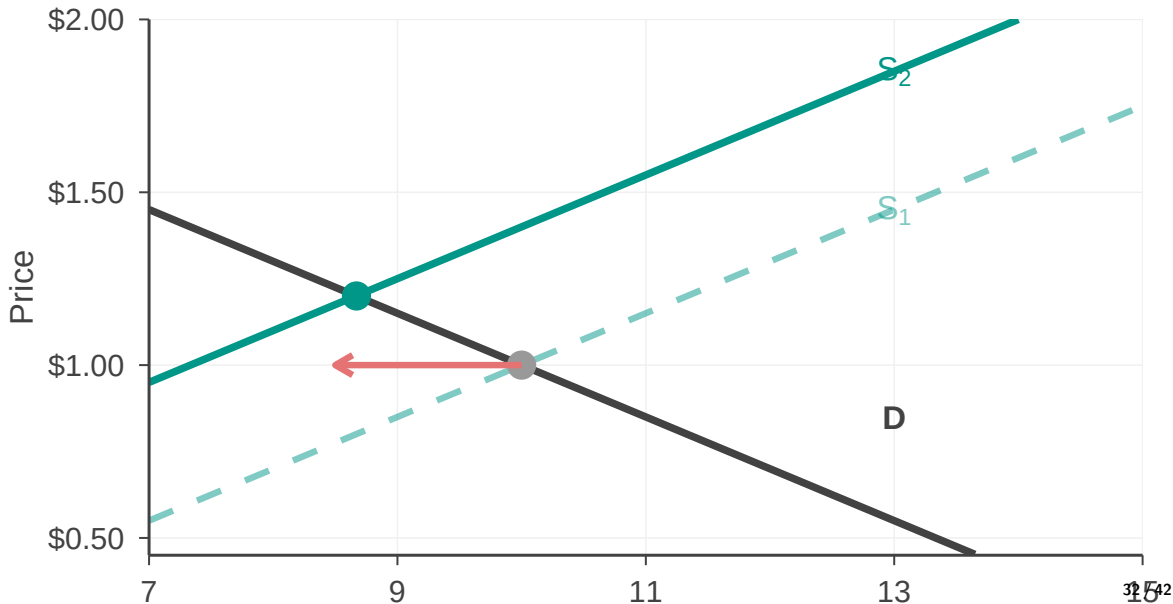
Knowledge Check: Shortage or Surplus?



When Demand Increases



When Supply Decreases



Practice: Predicting Market Changes

If the cost of wood falls, what happens to equilibrium price and quantity in the violin market?

Think about: 1. Which curve shifts? (Supply or demand?) 2. Which direction? (Left or right?)
3. What happens to P^* and Q^* ?

Practice: Predicting Market Changes

If the cost of wood falls, what happens to equilibrium price and quantity in the violin market?

Think about: 1. Which curve shifts? (Supply or demand?) 2. Which direction? (Left or right?) 3. What happens to P^* and Q^* ?

∴ .block ****Answer:**** Wood is an input $\rightarrow$ $\downarrow$ input prices $\rightarrow$ $\uparrow$ supply (shift right) $\rightarrow$ $\downarrow P^*$ and $\uparrow Q^*$ ∴

Simultaneous Shifts: Handle with Care

Example: iPhone launch

When Apple releases a new iPhone: 1. Consumer excitement → **demand increases** (curve shifts right) 2. Apple ramps up production → **supply increases** (curve shifts right)

What happens to equilibrium?

Both shifts push **quantity** in the same direction: - $\uparrow$ Demand $\rightarrow Q \uparrow$ - $\uparrow$ Supply $\rightarrow Q \uparrow$ - **Result: $Q \uparrow$ (certain)**

But shifts push **price** in opposite directions: - $\uparrow$ Demand $\rightarrow P \uparrow$ - $\uparrow$ Supply $\rightarrow P \downarrow$ - **Result: $P ?$ (ambiguous)**

General Rule

When both curves shift, if the effects on P or Q work in the **same direction** → certain change. If **opposite directions** → ambiguous (depends on relative magnitudes).

Interactive Demo: Open `interactive_simultaneous_shifts_v2.html` - use sliders to

Remember This?

How much does a 4oz bottle of pure vanilla extract cost?

- a) \$4**
- b) \$7**
- c) \$10**
- d) \$18**
- e) \$20**

The Vanilla Market: 2017



Madagascar vanilla beans

What happened:

- Vanilla prices: \$20/kg (2010) → \$500/kg (2017)
- **25× increase in 7 years**

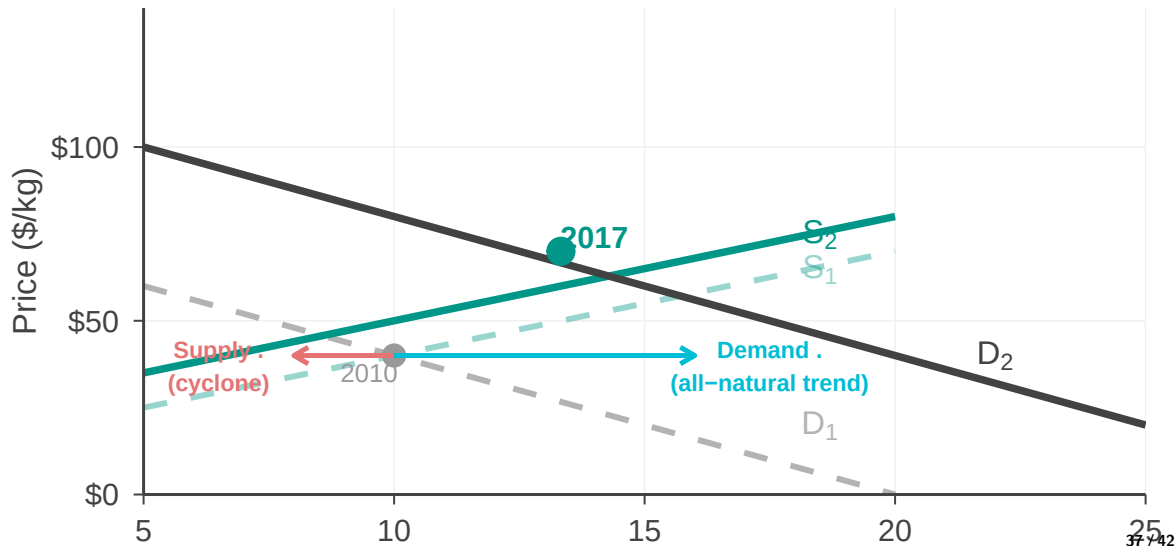
Why:

- ① Cyclone damaged 30% of Madagascar crop (supply ↓)
- ② “All-natural” food trend (demand ↑)
- ③ Major brands (Nestlé, Hershey’s) switched from synthetic to natural vanilla (demand ↑↑)



Vanilla: Supply and Demand Analysis

The Vanilla Crisis: Simultaneous Shifts



The Market Response

By 2019-2020, vanilla prices began falling:

Why?

- ① **High prices incentivized production** → farmers planted more vanilla → ↑ supply
- ② **Substitution** → some manufacturers switched back to synthetic vanillin → ↓ demand
- ③ **Recovery** → Madagascar crop recovered → ↑ supply

Classic Market Dynamics

High prices signal scarcity → encourage production and substitution → prices adjust back toward equilibrium

Key Takeaways

- ➊ **Demand curves slope down**, supply curves slope up
- ➋ **Movements** happen along curves when price changes; **shifts** happen when non-price factors change
- ➌ **TRIBE** shifts demand: Tastes, Related goods, Income, Buyers, Expectations
- ➍ **TONER** shifts supply: Technology, Other goods, Number of producers, Expectations, Resource costs
- ➎ **Equilibrium** balances Q_d and Q_s ; surpluses and shortages push price toward equilibrium
- ➏ **Price and quantity changes** depend on which curve shifts and in what direction

Final Knowledge Check

Garden gnomes regain popularity. What happens to equilibrium?

- a) $P \uparrow, Q \uparrow$
- b) $P \uparrow, Q \downarrow$
- c) $P \downarrow, Q \uparrow$
- d) $P \downarrow, Q \downarrow$

Final Knowledge Check

Garden gnomes regain popularity. What happens to equilibrium?

- a) $P \uparrow, Q \uparrow$
- b) $P \uparrow, Q \downarrow$
- c) $P \downarrow, Q \uparrow$
- d) $P \downarrow, Q \downarrow$

∴ .block ****Answer: (a)**** Tastes $\rightarrow$ demand shifts right $\rightarrow P \uparrow$ and $Q \uparrow$

This is the type of reasoning you'll use throughout the course! ∴

Next Time

Chapter 6: Elasticity

- How responsive is quantity to price changes?
- Why do some price changes barely affect quantity while others cause huge swings?
- What determines whether sellers or buyers bear the burden of a tax?

Before Next Class

- Review TRIBE and TONER
- Practice: For each news story you see this week, ask: “Which curve shifted? Which direction?”

Thank You

Questions?

ECON 1450: Principles of Microeconomics

University of Vermont